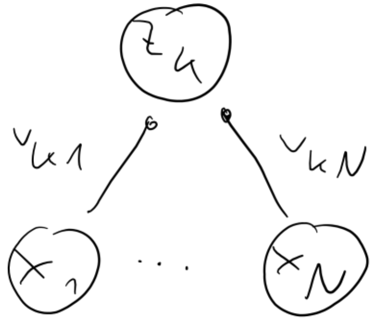
Tuning curves



$\vec{x} = (x_1, \dots, x_N)$

$\vec{V}_k = (V_{ki})_{1 \dots N}$

"What stimulus $\vec{x}^*$ is neuron z_k most responsive to?"

For static $\vec{x}(t) = \text{const}$,

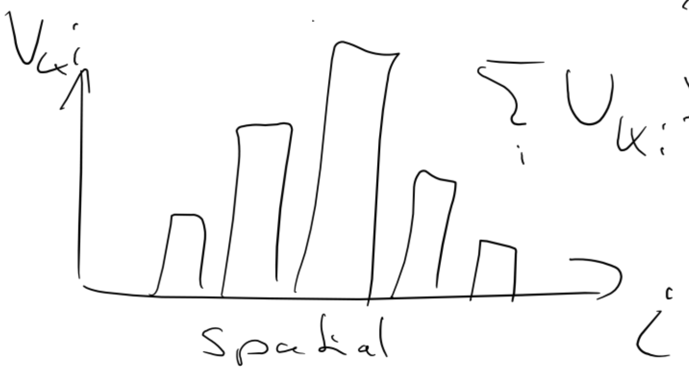
$$z_k \to f(\vec{V}_k \cdot \vec{x}) \quad \text{for } t \to \infty$$

· Since $f(\cdot)$ is (strictly) increasing (remember shape of response plot)

$$\vec{x}^* = \underset{\vec{x}}{\arg\max}\ \vec{V}_k \cdot \vec{x} = \underset{\vec{x}}{\arg\max}\ \|\vec{x}\| \|\vec{V}_k\| \cos(\sphericalangle(\vec{V}_k, \vec{x}))$$

↳ For fixed stimulus strength ($\|\vec{x}\| = \text{const.}$):

$$\vec{x}^* \parallel \vec{V}_k$$



$\sum_i V_{ki} x_i$

· often we think of $\vec{V}_k$ as a spatial kernel or a preferred input pattern:

"Neuron k is tuned to patterns $\vec{x} \parallel \vec{V}_k$"

Temporal dynamics

- For FF-networks, typically much simpler than for recurrent networks, (no feedback loops)
- Each neuron is a low-pass filter of its input $V_k(t)$

$$\left(\partial_t + \frac{1}{\tau_k}\right) z_k = \frac{1}{\tau_k} F\left(V_k(t) + b_k\right) =: f_k(t)$$

- Propagation of activity from layer to layer via Green's fct $G(t) = \Theta(t) e^{-t/\tau_k}$

$$z_k(t) = \int_{-\infty}^{\infty} G(t-t') f_k(t') \, dt'$$

- Simple example: $x \xrightarrow{V} z_1 \xrightarrow{W} z_2$

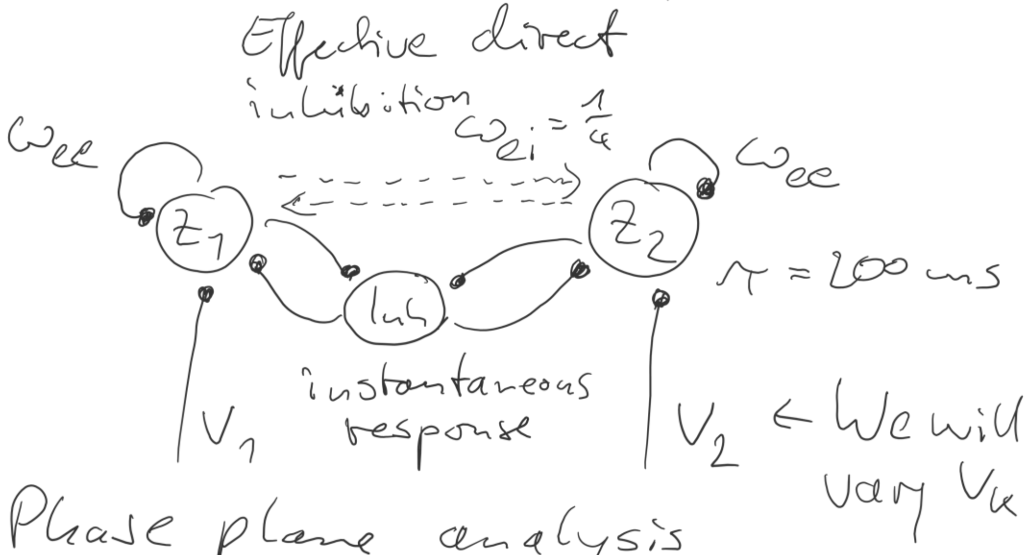
- $F_k = [\cdot]_+$ with $b_k = 0$
- $V, W > 0$
- $x(t) = I \delta(t)$

$$\hookrightarrow z_1(t) = \int_{-\infty}^{\infty} \Theta(t-t') e^{-\frac{t-t'}{\tau_1}} \frac{V}{\tau_1} I \delta(t') \, dt'$$

$$= \frac{I \cdot V}{\tau_1} \Theta(t) e^{-\frac{t}{\tau_1}}$$

$$\hookrightarrow z_2(t) = \int_{-\infty}^{\infty} \Theta(t-t') e^{-\frac{t-t'}{\tau_2}} \frac{W}{\tau_2} z_1(t') \, dt' = \ldots = \text{Diff. of exp.}$$

Competing excitatory populations



Inh. pop. senses the combined activity

$$\tau \dot{z}_k = -z_k + F\Big(w_{ee} z_k + w_{ei} \cdot \underbrace{F(z_1 + z_2)}_{} + V_k\Big)$$

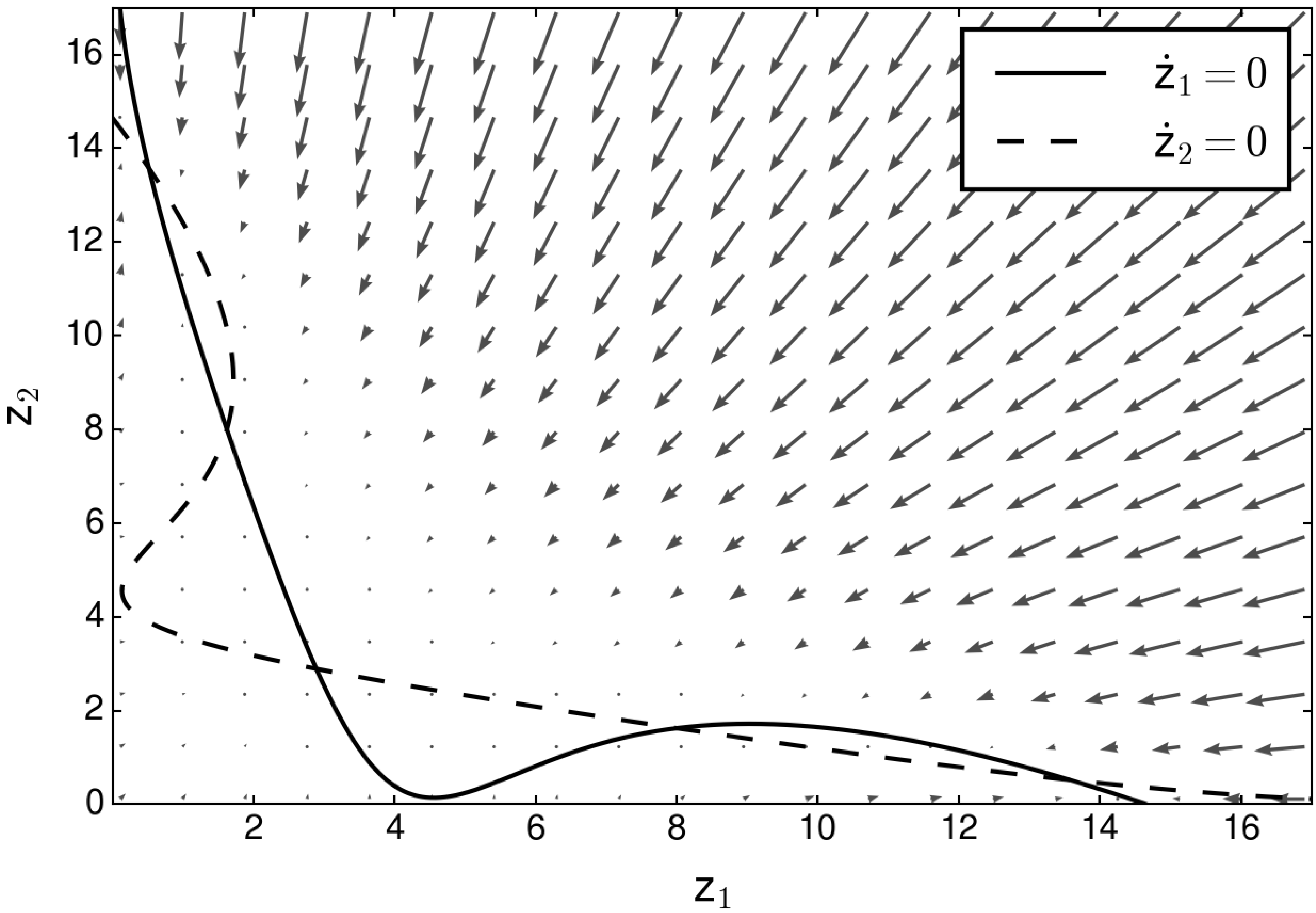
with $F(z) = \exp\big(\underset{\llcorner 1/4}{a} \cdot (z + \underset{\llcorner -2}{b})\big)$ for $k = 1, 2$

- Phase plane analysis
- Several scenarios
- $V_1 = V_2 = 4$ (weak input): Three stable fixed points

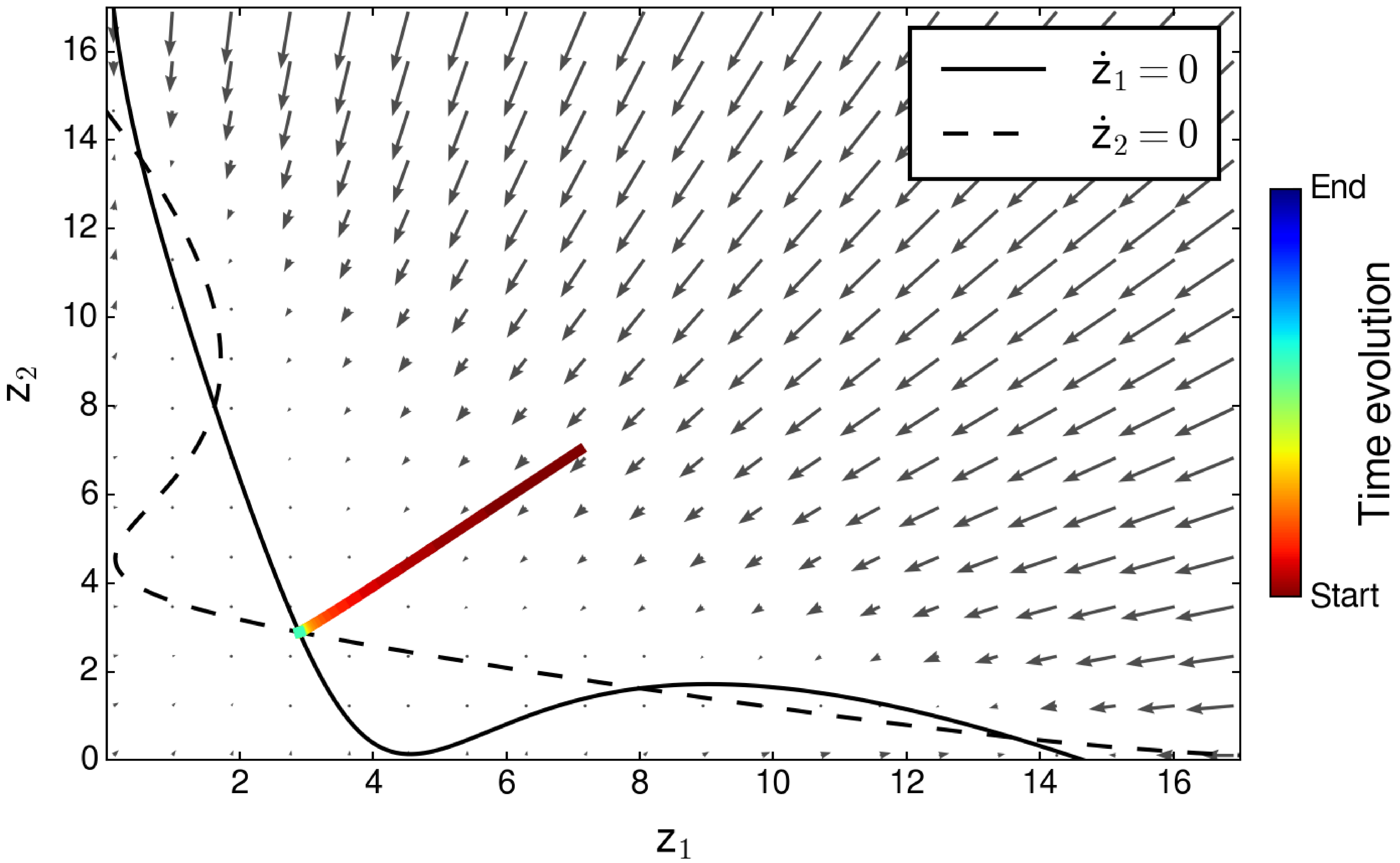
FP 1: Both populations fire at moderate frequency
FP 2 & 3: One population suppresses the other

- $|V_1 - V_2| > \delta$: only one stable fixed point
  → the winner takes almost all activity
- $V_1 = V_2 \geq 4.7$: Saddle-node bifurcation
  → even small differences in the initial state will lead to a winner

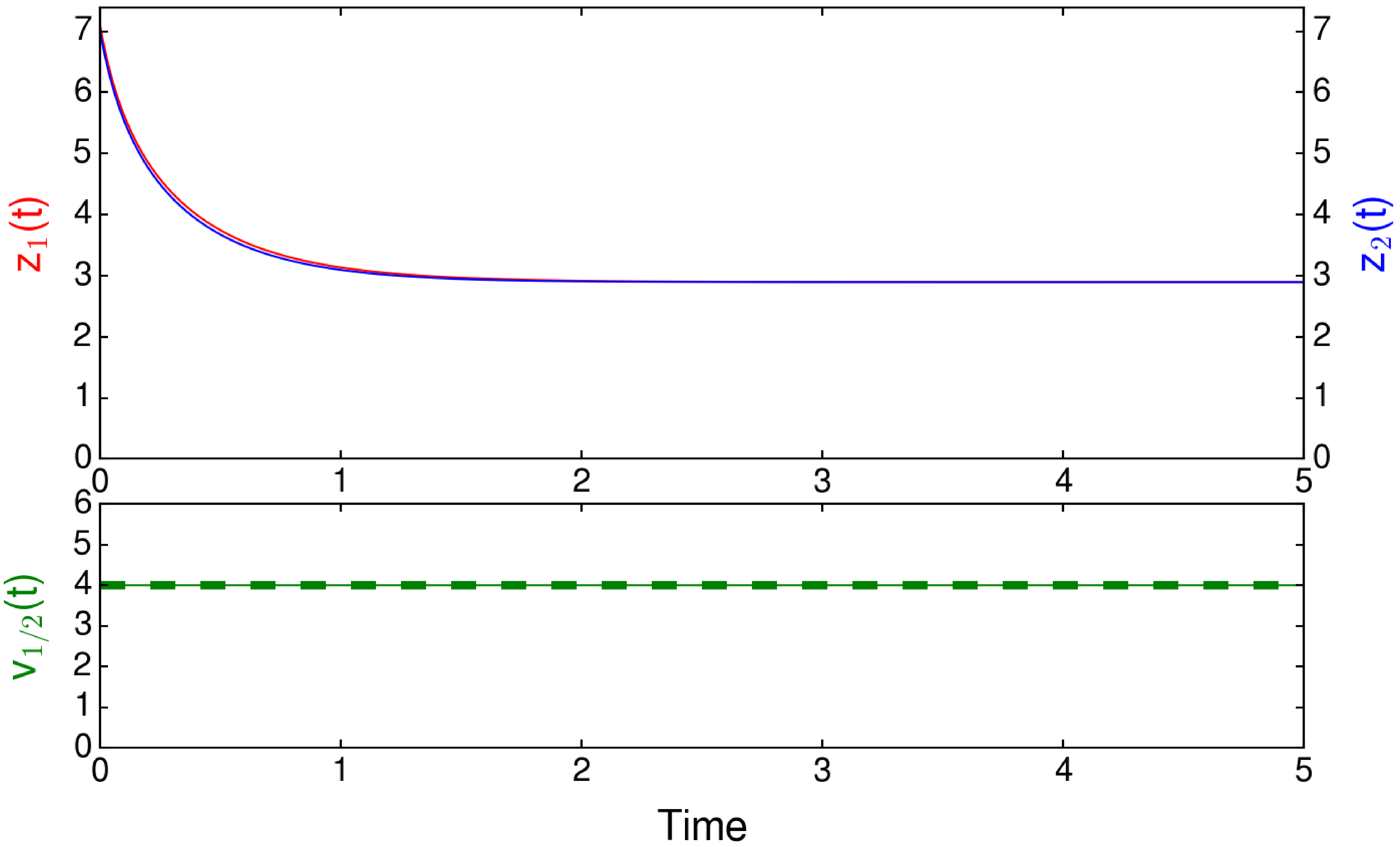
↳ Name: "Winner-takes-all network" (WTA)



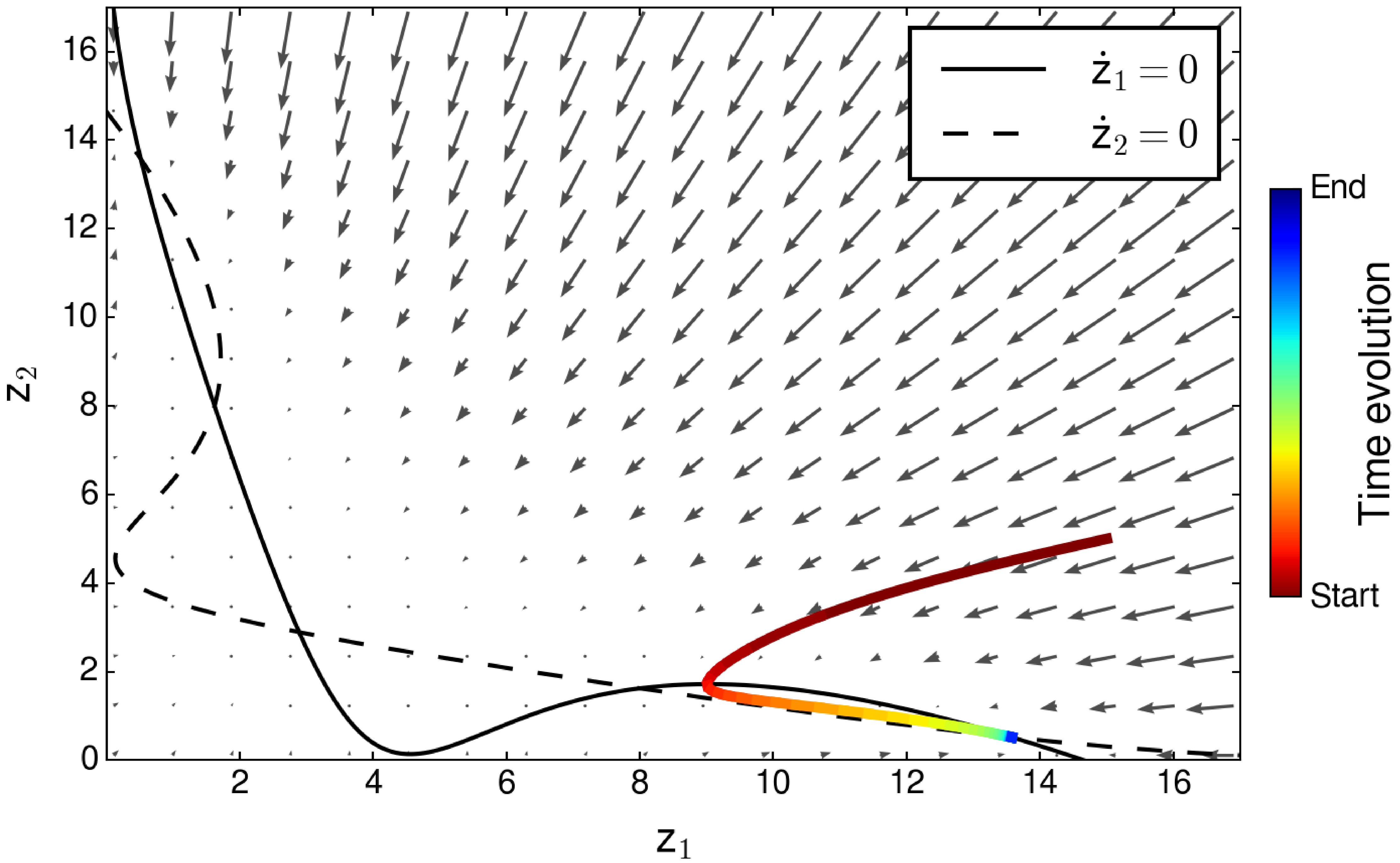
$\dot{z}_1 = 0$
$\dot{z}_2 = 0$
16
14
12
10
8
6
4
2
0
2
4
6
8
10
12
14
16
z_2
z_1



$\dot{z}_1 = 0$
$\dot{z}_2 = 0$
End
Start
Time evolution
z_2
z_1
0
2
4
6
8
10
12
14
16



$z_1(t)$
$z_2(t)$
$v_{1/2}(t)$
Time



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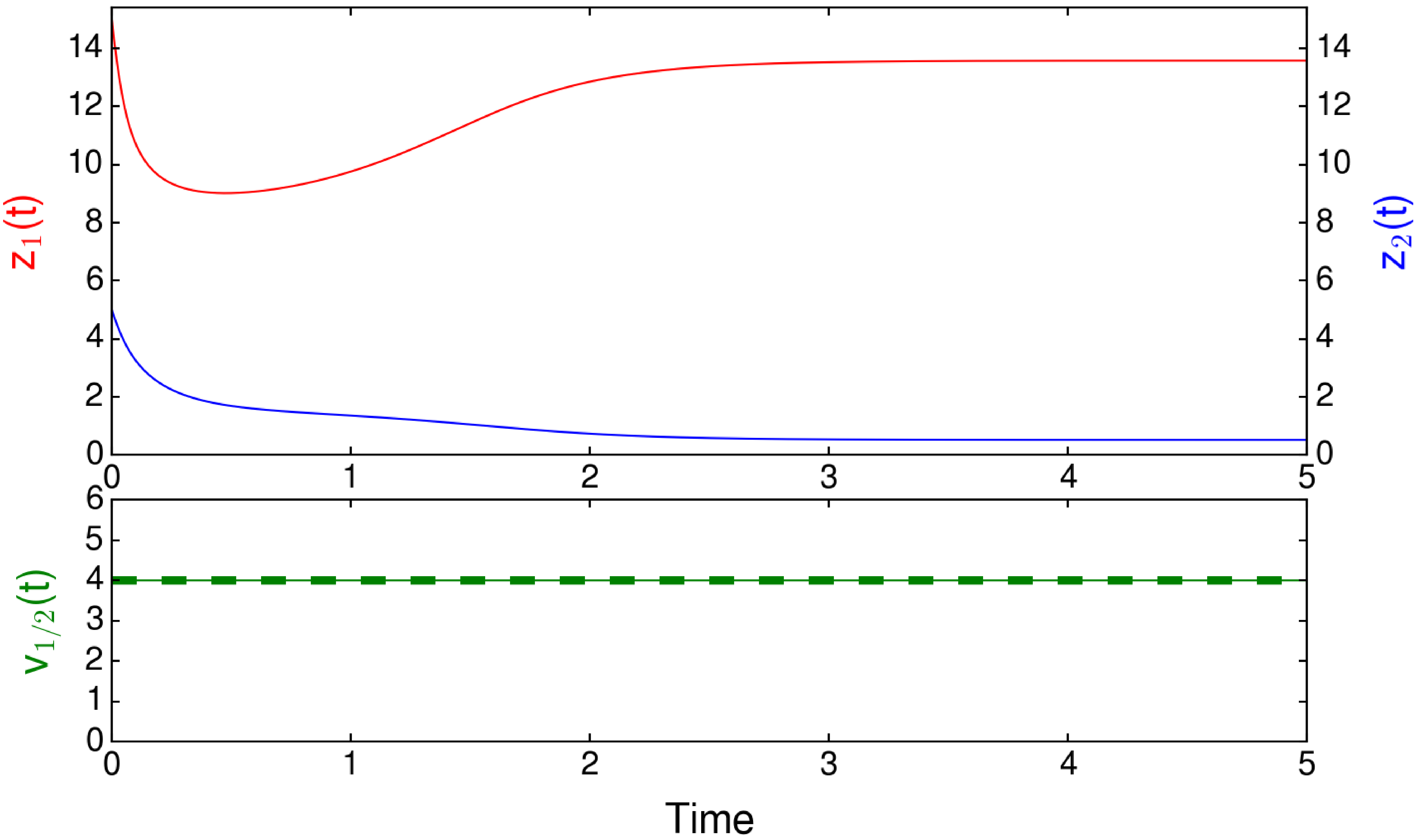

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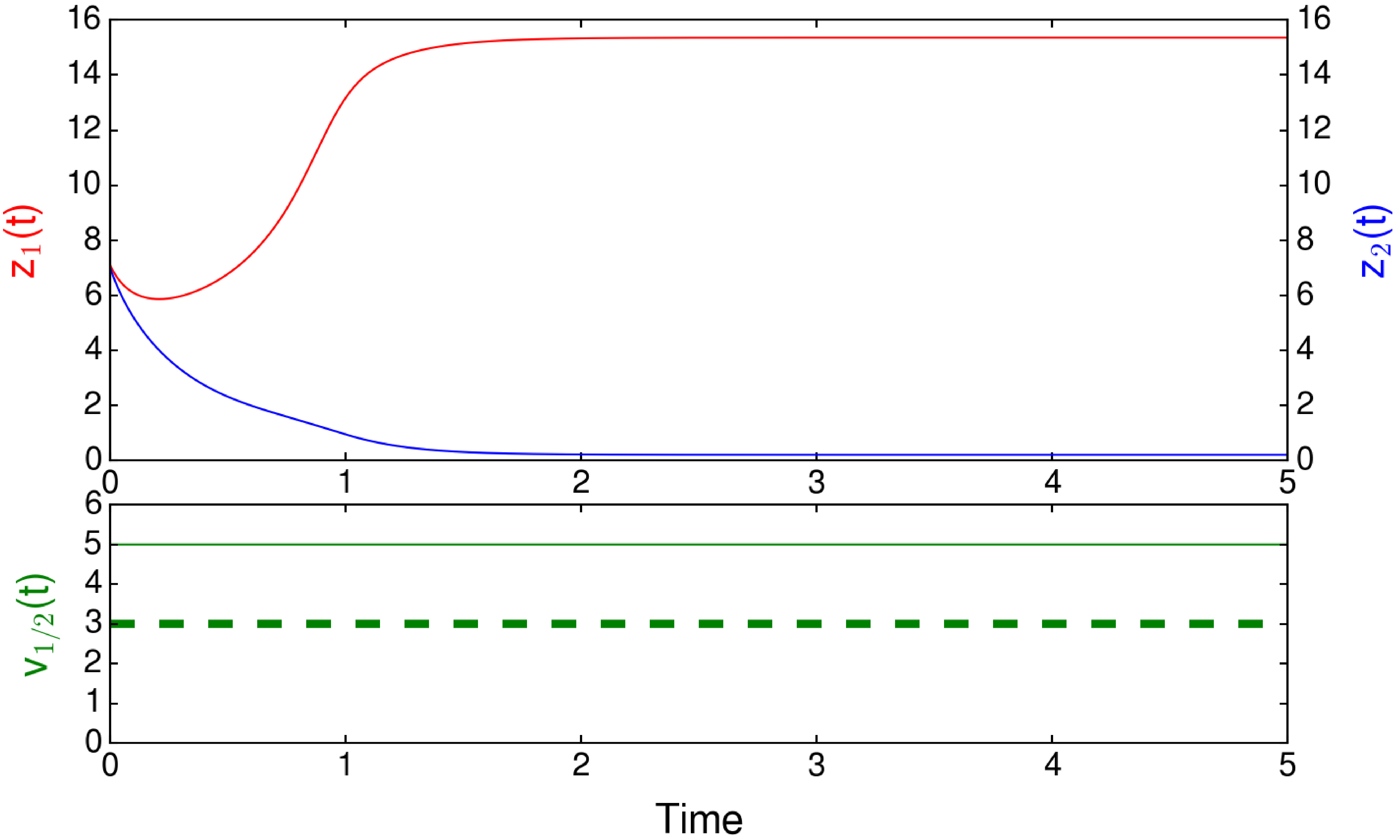
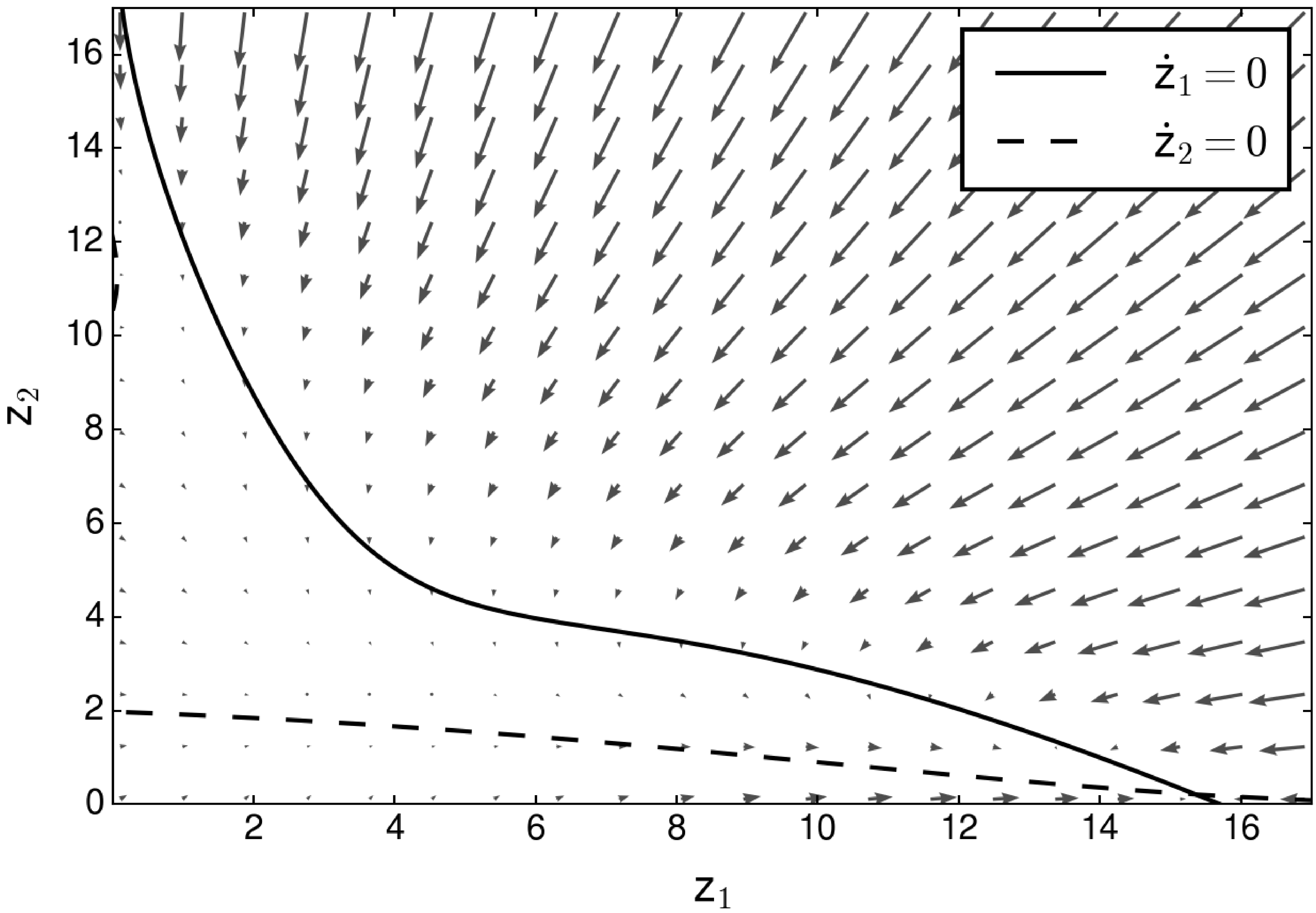
Time



$z_1(t)$
$z_2(t)$
$v_{1/2}(t)$
Time



$\dot{z}_1 = 0$
$\dot{z}_2 = 0$
z_2
z_1
0
2
4
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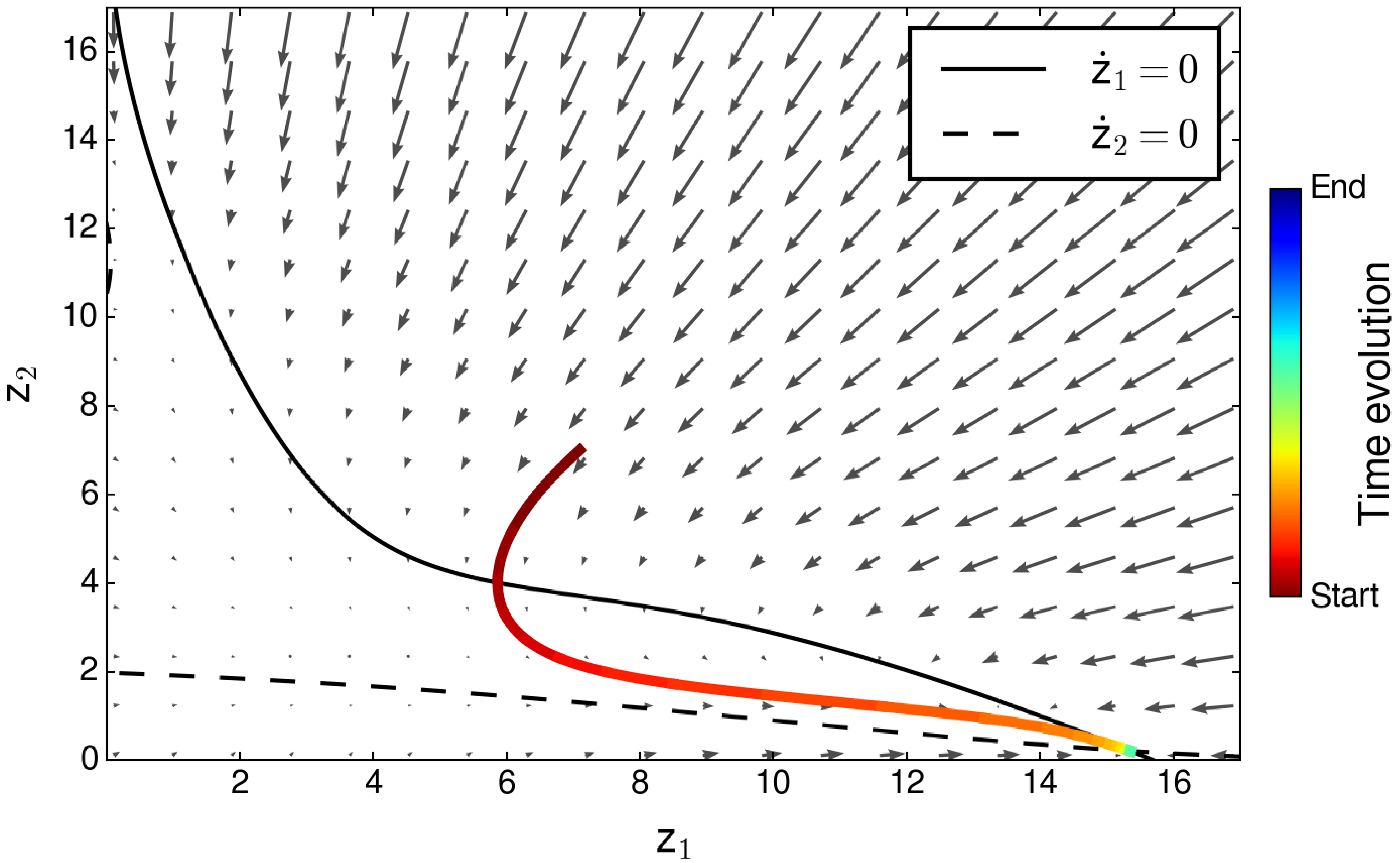
16



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Time evolution
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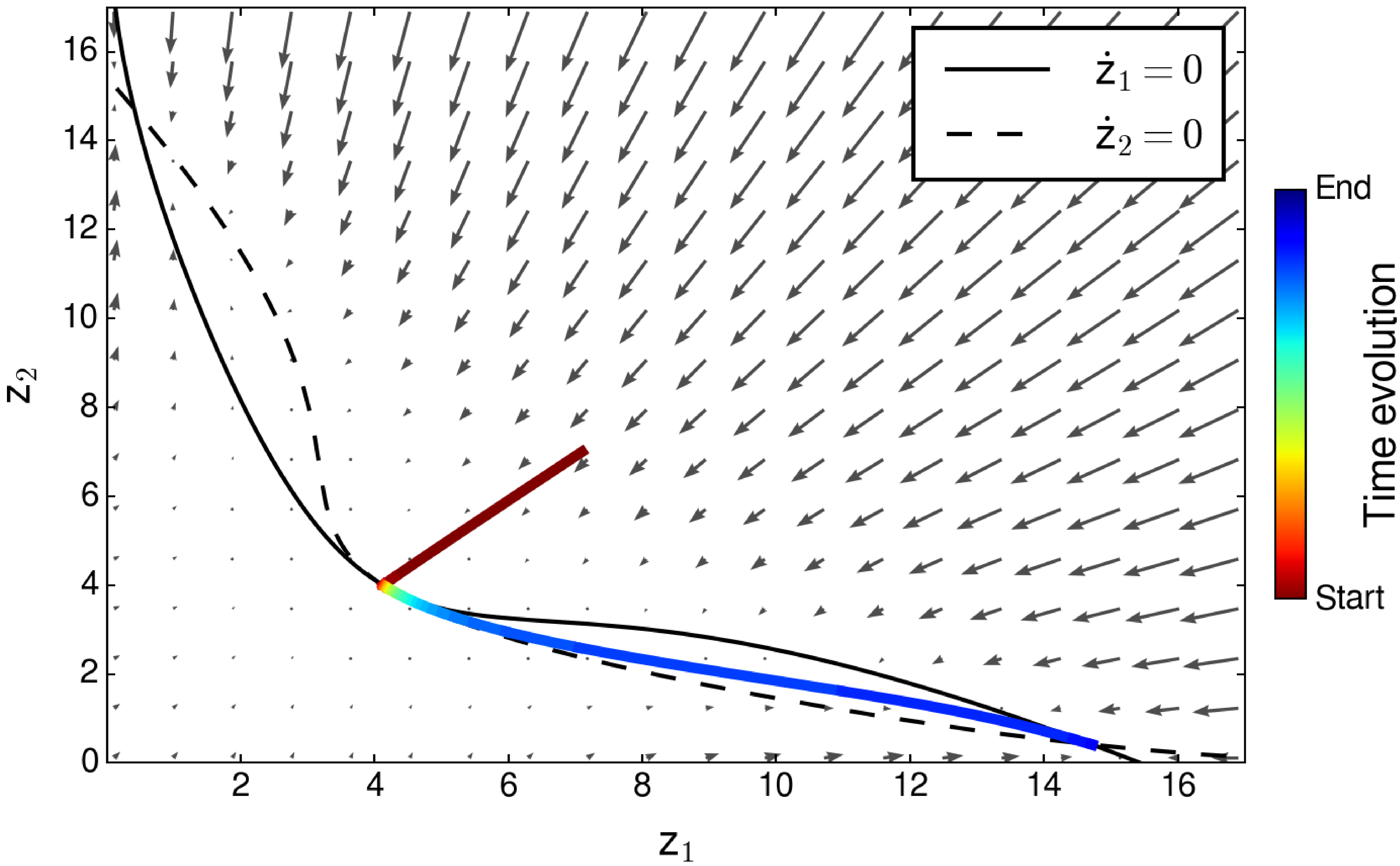

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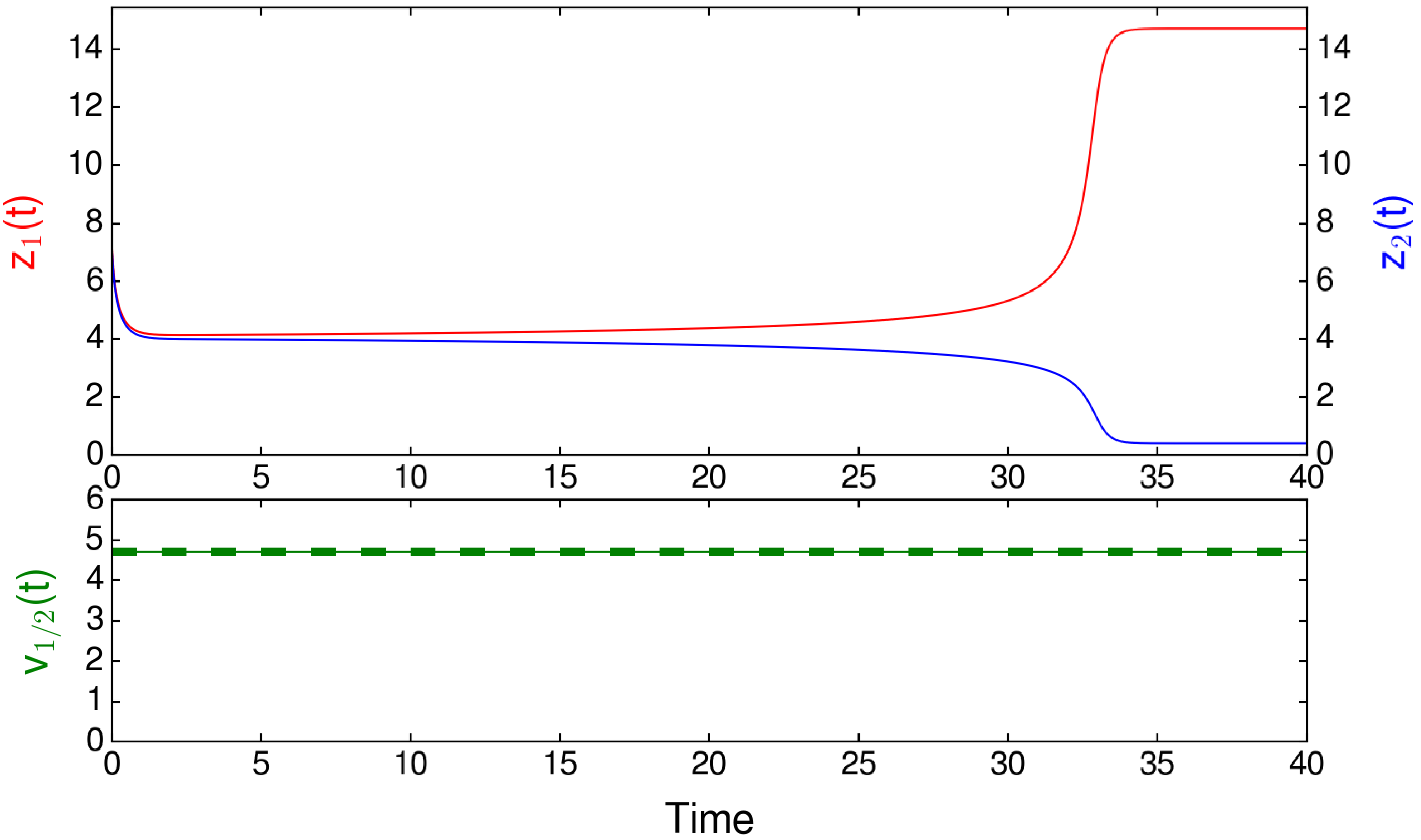
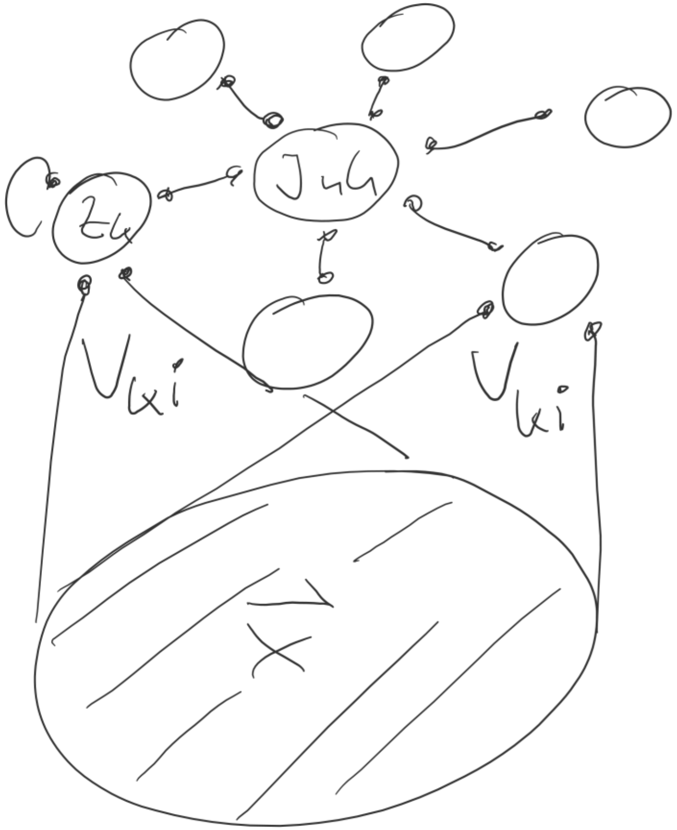


z_1(t)
z_2(t)
$v_{1/2}$(t)
Time

Input integration with multiple WTA-populations



WTA-competition: "Who matches best?"

Tuning to different patterns → $V_k = \sum_i V_{ki} x_i$

Presented input $\vec{x}$

- Reliable detection of inputs (Competition)
- Clear-cut output code (winner takes almost all activity).